

19.1 Omitting and adding regressors

The assumption $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ claims that the columns of $\mathbf{X}$ are the right ones. This section compares two ways of getting them wrong. *Underfitting* leaves out regressors that matter. *Overfitting* includes regressors that do not. This is a first look at the bias-variance trade-off of Part VI.

19.1.1 The setting

Throughout the section the truth is

$$E(\mathbf{Y}) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2, \quad \text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n,$$

where $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ has full column rank $p = p_1 + p_2 < n$. The *long* regression fits both blocks and gives $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$ and $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / (n - p)$. The *short* regression fits $\mathbf{X}_1$ alone and gives

$$\hat{\boldsymbol{\beta}}_1^S = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}, \quad s_1^2 = \frac{\|(\mathbf{I} - \mathbf{M}_1)\mathbf{Y}\|^2}{n - p_1},$$

where the superscript S marks the short fit and $\mathbf{M}_1$ projects onto $\mathcal{C}(\mathbf{X}_1)$. Two quantities recur. The first is $\boldsymbol{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$, the coefficients from regressing each column of $\mathbf{X}_2$ on $\mathbf{X}_1$, as in Proposition 5.7.2. The second is

$$\gamma = \frac{\|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\boldsymbol{\beta}_2\|^2}{\sigma^2} = \frac{\boldsymbol{\beta}_2^\top \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2 \boldsymbol{\beta}_2}{\sigma^2}. \quad (19.1.1)$$

Since $(\mathbf{M} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\boldsymbol{\beta}_2$, this is the noncentrality parameter Eq. 11.1.2 of the F test of $\boldsymbol{\beta}_2 = \mathbf{0}$ in the long model. It measures how far the true mean lies from the short model's column space, in units of σ^2 . It is zero iff $\boldsymbol{\beta}_2 = \mathbf{0}$, because $\mathbf{X}_2\boldsymbol{\beta}_2 \in \mathcal{C}(\mathbf{X}_1)$ forces $\boldsymbol{\beta}_2 = \mathbf{0}$ when $\mathbf{X}$ has full rank.

Both regressions are linked by an exact identity and by one covariance.

Lemma 19.1.1 — Short and long regressions

In this setting:

- (a) $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{Y}$ and $\hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^S - \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2$;
- (b) $\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\boldsymbol{\beta}}_2) = \mathbf{0}$ and $\text{Cov}(\hat{\boldsymbol{\beta}}_2) = \sigma^2 \mathbf{C}$, where $\mathbf{C} = (\mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2)^{-1}$.

Proof.

- (a) The first formula is Theorem 6.6.2(a) with the roles of the two blocks exchanged, and the second is Proposition 5.7.2(a). (b) Both estimators are linear in $\mathbf{Y}$. By Theorem 2.2.1,

$$\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\boldsymbol{\beta}}_2) = \sigma^2 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2 \mathbf{C} = \mathbf{0},$$

because $\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_1) = \mathbf{0}$. Similarly $\text{Cov}(\hat{\boldsymbol{\beta}}_2) = \sigma^2 \mathbf{C} \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1)^2 \mathbf{X}_2 \mathbf{C} = \sigma^2 \mathbf{C}$.

19.1.2 Underfitting

Theorem 19.1.2 — Omitted regressors

Fit the short model when the long model is true.

- (a) $E(\hat{\beta}_1^S) = \beta_1 + \Pi\beta_2$ and $\text{Cov}(\hat{\beta}_1^S) = \sigma^2(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$. The bias $\Pi\beta_2$ vanishes iff $\mathbf{X}_1^\top \mathbf{X}_2\beta_2 = \mathbf{0}$, and it vanishes for every β_2 iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.
- (b) The fitted values have mean $\mathbf{X}\beta - (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$. The residuals have mean $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$ and covariance $\sigma^2(\mathbf{I} - \mathbf{M}_1)$, as if the short model were right.
- (c) $E(s_1^2) = \sigma^2(1 + \gamma/(n - p_1))$, so s_1^2 overestimates σ^2 unless $\beta_2 = \mathbf{0}$.
- (d) At a new point with regressors $(\mathbf{x}_{10}, \mathbf{x}_{20})$, the short prediction $\mathbf{x}_{10}^\top \hat{\beta}_1^S$ has bias $(\Pi^\top \mathbf{x}_{10} - \mathbf{x}_{20})^\top \beta_2$ for the mean $\mathbf{x}_{10}^\top \beta_1 + \mathbf{x}_{20}^\top \beta_2$.

Proof.

- (a) The mean is Proposition 5.7.2(b). The covariance is Theorem 5.5.1(b) for the model matrix $\mathbf{X}_1$, which needs only $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$. Since $(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$ is nonsingular, $\Pi\beta_2 = \mathbf{0}$ iff $\mathbf{X}_1^\top \mathbf{X}_2\beta_2 = \mathbf{0}$, and this holds for all β_2 iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.
- (b) $E(\mathbf{M}_1\mathbf{Y}) = \mathbf{X}_1\beta_1 + \mathbf{M}_1\mathbf{X}_2\beta_2 = \mathbf{X}\beta - (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$, and $E((\mathbf{I} - \mathbf{M}_1)\mathbf{Y}) = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$ because $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_1 = \mathbf{0}$. The covariance is $(\mathbf{I} - \mathbf{M}_1)\sigma^2\mathbf{I}(\mathbf{I} - \mathbf{M}_1) = \sigma^2(\mathbf{I} - \mathbf{M}_1)$.
- (c) By Eq. 2.3.1 with $\mathbf{M}_1$ of rank p_1 and mean $\mathbf{X}\beta$, $E\|(\mathbf{I} - \mathbf{M}_1)\mathbf{Y}\|^2 = \sigma^2(n - p_1) + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2\|^2 = \sigma^2(n - p_1 + \gamma)$.
- (d) Take expectations in $\mathbf{x}_{10}^\top \hat{\beta}_1^S$ using (a), and subtract the true mean.

Nothing in the second moments signals the omission: the covariances of $\hat{\beta}_1^S$ and of the residuals are exactly what the short model claims. The damage is in the first moments. The estimator is centred at the wrong place, and the residuals carry the systematic component $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$ that added-variable plots (Theorem 6.6.2) and lack-of-fit tests (Theorem 14.6.2) look for. Because s_1^2 is inflated, the reported variances are on average too large, yet as n grows they shrink while the bias does not, so the coverage of intervals for β_1 tends to zero. Nothing requires a coefficient to be omitted: for any true mean $\mathbf{X}_1\beta_1 + \delta$, such as a curved mean fitted by a line, (a)–(c) hold with $\mathbf{X}_2\beta_2$ replaced by δ (Proposition 5.5.7(a)).

19.1.3 Overfitting

Theorem 19.1.3 — Irrelevant regressors

Suppose $\beta_2 = \mathbf{0}$, so the short model is correct, but the long model is fitted.

- (a) $E(\hat{\beta}_1) = \beta_1$, $E(\hat{\beta}_2) = \mathbf{0}$ and $E(s^2) = \sigma^2$.
- (b) $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S) + \sigma^2 \Pi C \Pi^\top$. Hence $\text{Var}(\mathbf{a}^\top \hat{\beta}_1) \geq \text{Var}(\mathbf{a}^\top \hat{\beta}_1^S)$ for every $\mathbf{a} \in \mathbb{R}^{p_1}$, with equality iff $\Pi^\top \mathbf{a} = \mathbf{0}$, and $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S)$ iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.
- (c) At a new point $\mathbf{x}_0 = (\mathbf{x}_{10}^\top, \mathbf{x}_{20}^\top)^\top$,

$$\begin{aligned} \text{Var}(\mathbf{x}_0^\top \hat{\beta}) &= \text{Var}(\mathbf{x}_{10}^\top \hat{\beta}_1^S) \\ &\quad + \sigma^2 (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10})^\top \mathbf{C} (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10}), \end{aligned}$$

and summed over the design points $E\|\mathbf{X}\hat{\beta} - \mathbf{X}\beta\|^2 = \sigma^2 p$, against $\sigma^2 p_1$ for the short fit.

Proof.

- (a) The long model is correct (with $\beta_2 = \mathbf{0}$), so Theorem 5.5.1(a) and Theorem 5.6.1 apply.
- (b) By Lemma 19.1.1(a), $\hat{\beta}_1 = \hat{\beta}_1^S - \Pi \hat{\beta}_2$ with the two terms uncorrelated, so $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S) + \Pi \text{Cov}(\hat{\beta}_2) \Pi^\top$, and Lemma 19.1.1(b) gives $\text{Cov}(\hat{\beta}_2)$. Because $\mathbf{C}$ is positive definite, $\mathbf{a}^\top \Pi C \Pi^\top \mathbf{a} = 0$ iff $\Pi^\top \mathbf{a} = \mathbf{0}$, and $\Pi C \Pi^\top = \mathbf{0}$ iff $\Pi = \mathbf{0}$, iff $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$.
- (c) Write $\mathbf{x}_0^\top \hat{\beta} = \mathbf{x}_{10}^\top \hat{\beta}_1^S + (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10})^\top \hat{\beta}_2$ using Lemma 19.1.1(a), and use the zero covariance of (b) of that lemma. For the sum, $\mathbf{X}\hat{\beta} = \mathbf{M}\mathbf{Y}$ is unbiased for $\mathbf{X}\beta$ with covariance $\sigma^2 \mathbf{M}$, so $E\|\mathbf{M}\mathbf{Y} - \mathbf{X}\beta\|^2 = \sigma^2 \text{tr } \mathbf{M} = \sigma^2 p$. The short fit $\mathbf{M}_1 \mathbf{Y}$ is also unbiased, and its trace is p_1 .

■

An irrelevant regressor costs no bias, only variance, and the price grows with its correlation with the relevant columns: for one extra column the variance of a coefficient is multiplied by $1/(1 - r^2)$, with r the partial correlation of the two regressors (Eq. 6.6.2).

19.1.4 Bias against variance

When β_2 is small but not zero, the short regression is biased but less variable. Which is better? Compare the two estimators of β by their mean squared error matrices, writing $\hat{\beta}^S = ((\hat{\beta}_1^S)^\top, \mathbf{0}^\top)^\top$ for the short estimator, which sets β_2 to zero.

Lemma 19.1.4 — A rank-one comparison

Let $\mathbf{C}$ be a positive definite $q \times q$ matrix, $\mathbf{b} \in \mathbb{R}^q$ and $c > 0$. Then $c\mathbf{C} - \mathbf{b}\mathbf{b}^\top$ is nonnegative definite iff $\mathbf{b}^\top \mathbf{C}^{-1} \mathbf{b} \leq c$.

Proof. By Lemma 12.2.3 with $\mathbf{W} = \mathbf{C}$ and $\mathbf{u} = \mathbf{b}$, the inequality $\mathbf{b}^\top \mathbf{C}^{-1} \mathbf{b} \leq c$ holds iff $(\mathbf{a}^\top \mathbf{b})^2 \leq c \mathbf{a}^\top \mathbf{C} \mathbf{a}$ for every $\mathbf{a}$, that is, iff $\mathbf{a}^\top (c\mathbf{C} - \mathbf{b}\mathbf{b}^\top) \mathbf{a} \geq 0$ for every $\mathbf{a}$.

■

Theorem 19.1.5 — When the short regression wins

In the setting of this section, let $\mathbf{G} = \begin{pmatrix} -\mathbf{\Pi} \\ \mathbf{I}_{p_2} \end{pmatrix}$, a $p \times p_2$ matrix.

- (a) The mean squared error matrix of $\hat{\boldsymbol{\beta}}^S$ is

$$E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})^\top] = \text{Cov}(\hat{\boldsymbol{\beta}}) - \mathbf{G}(\sigma^2\mathbf{C} - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{G}^\top.$$

- (b) $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}^S - \boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 \leq \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ for every $\boldsymbol{\lambda} \in \mathbb{R}^p$ iff $\gamma \leq 1$. If $\gamma > 1$, the reverse strict inequality holds for some $\boldsymbol{\lambda}$.
- (c) Over the design points, $E\|\mathbf{X}\hat{\boldsymbol{\beta}}^S - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2(p_1 + \gamma)$, against $\sigma^2 p$ for the long fit. The short fit is better there iff $\gamma < p_2$.

Proof.

- (a) By Lemma 19.1.1(a), $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^S + \mathbf{G}\hat{\boldsymbol{\beta}}_2$, and $\hat{\boldsymbol{\beta}}^S$ is uncorrelated with $\hat{\boldsymbol{\beta}}_2$ by part (b) of the lemma. Since $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ and $E(\hat{\boldsymbol{\beta}}_2) = \boldsymbol{\beta}_2$, $E(\hat{\boldsymbol{\beta}}^S) - \boldsymbol{\beta} = -\mathbf{G}\boldsymbol{\beta}_2$. Expand $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta} + \mathbf{G}\hat{\boldsymbol{\beta}}_2)(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta} + \mathbf{G}\hat{\boldsymbol{\beta}}_2)^\top]$. The cross term is $E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})\hat{\boldsymbol{\beta}}_2^\top] = \text{Cov}(\hat{\boldsymbol{\beta}}^S, \hat{\boldsymbol{\beta}}_2) + (E\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})\boldsymbol{\beta}_2^\top = -\mathbf{G}\boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$, and $E(\hat{\boldsymbol{\beta}}_2\hat{\boldsymbol{\beta}}_2^\top) = \sigma^2\mathbf{C} + \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$. So

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})^\top] + \mathbf{G}(\sigma^2\mathbf{C} + \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top - 2\boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{G}^\top.$$

- (b) Put $\mathbf{A} = \sigma^2\mathbf{C} - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$. The claimed inequality for all $\boldsymbol{\lambda}$ says that $\mathbf{GAG}^\top$ is nonnegative definite. The matrix $\mathbf{L} = [\mathbf{0}, \mathbf{I}_{p_2}]$ satisfies $\mathbf{LG} = \mathbf{I}$, so $\mathbf{A} = \mathbf{L}(\mathbf{GAG}^\top)\mathbf{L}^\top$. Hence $\mathbf{GAG}^\top$ is nonnegative definite iff $\mathbf{A}$ is, and by Lemma 19.1.4 iff $\boldsymbol{\beta}_2^\top \mathbf{C}^{-1} \boldsymbol{\beta}_2 \leq \sigma^2$, which is $\gamma \leq 1$. If $\gamma > 1$, the same lemma gives $\mathbf{a}$ with $\mathbf{a}^\top \mathbf{A} \mathbf{a} < 0$, and $\boldsymbol{\lambda} = \mathbf{L}^\top \mathbf{a}$ satisfies $\mathbf{G}^\top \boldsymbol{\lambda} = \mathbf{a}$, so $\boldsymbol{\lambda}^\top \mathbf{GAG}^\top \boldsymbol{\lambda} < 0$.
- (c) $\mathbf{X}\hat{\boldsymbol{\beta}}^S = \mathbf{M}_1\mathbf{Y}$, whose mean is $\mathbf{M}_1\mathbf{X}\boldsymbol{\beta}$ and covariance $\sigma^2\mathbf{M}_1$. So $E\|\mathbf{M}_1\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2 p_1 + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2(p_1 + \gamma)$. The long fit is Theorem 19.1.3(c), whose proof did not use $\boldsymbol{\beta}_2 = \mathbf{0}$ for this part. ■

The threshold in (b) depends only on the noncentrality, not on the design or on n . It covers all linear functions at once; for a particular function the short estimator can win for larger γ , as (c) shows. Since γ is unknown, neither criterion applies directly. But the F statistic for $\boldsymbol{\beta}_2 = \mathbf{0}$ depends on the parameters only through γ (Theorem 11.1.1), so a small F is evidence that γ is small. Turning this into a procedure, and paying for having chosen the model from the data, is the subject of Chapter 29.

Example. (A biased estimator that wins)

Take $n = 30$ observations of two centred regressors with unit sample variances and sample correlation $r = 0.9$, an intercept, $\sigma = 1$ and $\beta_2 = 0.301$, chosen so that $\gamma = 0.5$. Here $\mathbf{\Pi}$ has slope entry r , so the short slope on x_1 has bias $r\beta_2 = 0.271$. Its standard deviation is 0.186, against 0.426 for the long slope, and the ratio of mean squared error to long variance is 0.595. A simulation with 200000 data sets gives 0.594.

For two standardized regressors the ratio has a closed form ([Exercise \(B2\)](#)):

$$\frac{E(\hat{\beta}_1^S - \beta_1)^2}{\text{Var}(\hat{\beta}_1)} = 1 - r^2 + r^2\gamma = 1 + r^2(\gamma - 1), \quad (19.1.2)$$

which is $1 - 0.81 + 0.81 \times 0.5 = 0.595$. The short slope wins iff $\gamma < 1$, as [Theorem 19.1.5\(b\)](#) requires, and the stakes, gain or loss, are proportional to r^2 . This is panel (a) of [Figure 19.1.1](#). The short model's s_1^2 has mean $1 + \gamma / (n - 2) = 1.0179$, a bias too small to notice.

```
import numpy as np
rng = np.random.default_rng(1901)
n, r, sigma = 30, 0.9, 1.0

# two centred, unit-variance regressors with sample correlation exactly r:
# orthonormalize (1, z1, z2) and keep the last two columns, which are centred
Z = np.linalg.qr(np.column_stack([np.ones(n), rng.normal(size=(n, 2))]))[0][:, 1:]
Z *= np.sqrt(n - 1)
x1 = Z[:, 0]
x2 = r * Z[:, 0] + np.sqrt(1 - r**2) * Z[:, 1]
X1 = np.column_stack([np.ones(n), x1])           # the short model
X = np.column_stack([X1, x2])                   # the long model

M1 = X1 @ np.linalg.solve(X1.T @ X1, X1.T)
S22_1 = x2 @ (np.eye(n) - M1) @ x2               # ||(I - M1) x2||^2
beta2 = np.sqrt(0.5 * sigma**2 / S22_1)          # chosen so that gamma = 0.5
gamma = beta2**2 * S22_1 / sigma**2

var_long = sigma**2 * np.linalg.inv(X.T @ X)[1, 1] # Var of the long slope on x1
Pi = np.linalg.solve(X1.T @ X1, X1.T @ x2)[1]    # slope of x2 on x1
var_short = sigma**2 * np.linalg.inv(X1.T @ X1)[1, 1]
mse_short = var_short + (Pi * beta2) ** 2          # variance + squared bias
print(f"gamma = {gamma:.3f}, MSE(short)/Var(long) = {mse_short / var_long:.4f}")
print(f"formula 1 - r^2 + r^2 gamma = {1 - r**2 + r**2 * gamma:.4f}")
```

[Run this code online](#)

```
reps = 200_000
beta = np.array([1.0, 1.0, beta2])
Y = X @ beta + sigma * rng.normal(size=(reps, n))
b_long = np.linalg.solve(X.T @ X, X.T @ Y.T)[1]
b_short = np.linalg.solve(X1.T @ X1, X1.T @ Y.T)[1]
print("simulated MSE ratio:", np.mean((b_short - 1) ** 2) / np.mean((b_long - 1) ** 2))
```

[Run this code online](#)

Example. (Choosing a polynomial degree)

Part (c) of [Theorem 19.1.5](#) applies to any nested sequence of models. Take $n = 40$ equally spaced points on $[-1, 1]$, the mean $\mu(x) = e^x \sin 3x$, which is not a polynomial, and $\sigma = 0.25$. A polynomial of degree d has $p = d + 1$ columns, and its risk at the design points is $\sigma^2(d + 1) + \|(\mathbf{I} - \mathbf{M}_d)\boldsymbol{\mu}\|^2$. Panel (b) of [Figure 19.1.1](#) shows the two terms. In units of σ^2 the risk is 161.41 for a straight line, falls to 5.52 at degree 4, and climbs back to 11.00 at degree 10, where the fit is almost unbiased and pays only for its

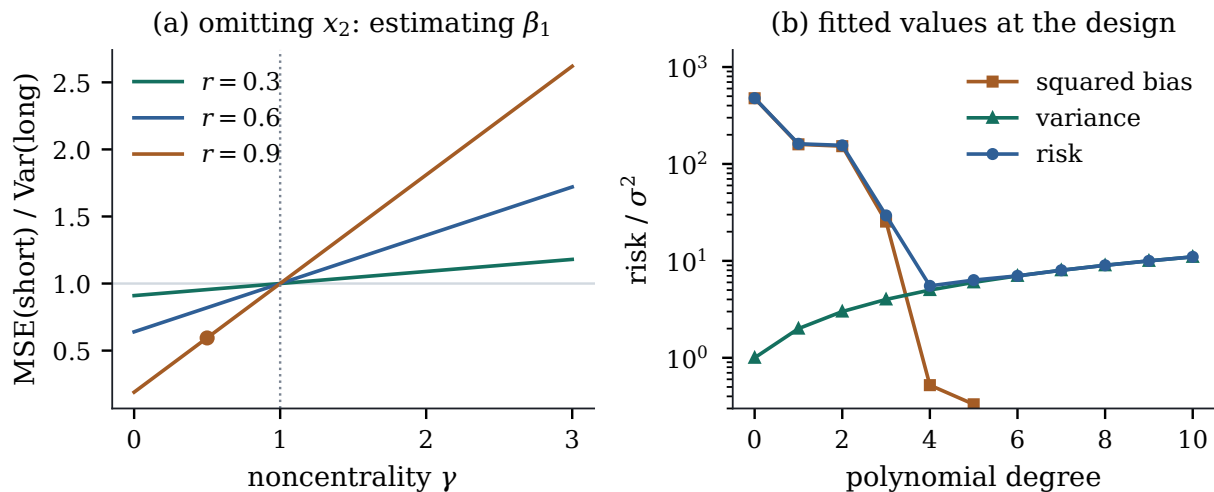


Figure 19.1.1: (a) Mean squared error of the short slope relative to the variance of the long slope, Eq. 19.1.2, for three correlations. All lines cross at $\gamma = 1$; the point is the simulation of [Example \(A biased estimator that wins\)](#). (b) Squared bias, variance and their sum for polynomial fits of increasing degree ([Example \(Choosing a polynomial degree\)](#)), in units of σ^2 and on a logarithmic scale.

variance. [Exercise \(B3\)](#) shows that adding one term lowers the risk iff that term's own noncentrality exceeds 1.

Idea. Omitting a regressor that matters costs bias; keeping one that does not costs variance, and for $\gamma < 1$ the bias is the cheaper. This does not make the biased coefficient *interpretable*: it still estimates $\beta_1 + \Pi\beta_2$, and whether that answers the question is a matter for [Chapter 25](#).

19.1.5 Exercises

19.1.5.1 A. Check your understanding

Exercise A1.

A straight line is fitted at $x = 0, 1, 2, 3$ when the true mean is $\beta_0 + \beta_1 x + \beta_2 x^2$. Find the biases of the fitted intercept and slope.

Exercise A2.

True or false: if the omitted regressors are orthogonal to the included ones ($\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$), the short regression is harmless, including its estimate of σ^2 .

19.1.5.2 B. Practice

Exercise B1.

In [Exercise \(A1\)](#), show that $\|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\|^2 = 4$. For which values of β_2/σ does the

straight line have a smaller mean squared error matrix than the quadratic fit, in the sense of [Theorem 19.1.5\(b\)](#)? For which values is it better at the four design points?

Exercise B2.

Derive [Eq. 19.1.2](#). Take an intercept and two centred regressors with $\|\mathbf{x}_1\|^2 = \|\mathbf{x}_2\|^2 = S$ and sample correlation r . Show that $\text{Var}(\hat{\beta}_1) = \sigma^2/(S(1-r^2))$, that the short slope has variance σ^2/S and bias $r\beta_2$, and that $\gamma = \beta_2^2 S(1-r^2)/\sigma^2$.

Solution. The long variance is [Eq. 6.6.2](#) with $R_1^2 = r^2$. The short slope is $\mathbf{x}_1^\top \mathbf{Y}/S$, with variance σ^2/S and mean $\beta_1 + \beta_2 \mathbf{x}_1^\top \mathbf{x}_2/S = \beta_1 + r\beta_2$. The residual of $\mathbf{x}_2$ after regression on $(\mathbf{1}, \mathbf{x}_1)$ has squared length $S(1-r^2)$, which gives γ . Then $(\sigma^2/S + r^2\beta_2^2)S(1-r^2)/\sigma^2 = 1 - r^2 + r^2\gamma$.

Exercise B3.

Let $\mathcal{C}(\mathbf{X}_1) \subset \mathcal{C}(\mathbf{X}_2) \subset \dots$ be nested model spaces with dimensions $p_1 < p_2 < \dots$, and let $\boldsymbol{\mu} = E(\mathbf{Y})$ be arbitrary. Show that the risk $E\|\mathbf{M}_k \mathbf{Y} - \boldsymbol{\mu}\|^2$ of model k exceeds that of model $k+1$ iff $\gamma_k = \|(\mathbf{M}_{k+1} - \mathbf{M}_k)\boldsymbol{\mu}\|^2/\sigma^2 > p_{k+1} - p_k$.

Solution. As in [Theorem 19.1.5\(c\)](#), the risk of model k is $\sigma^2 p_k + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2$. By [Theorem 6.5.1](#), $\mathbf{I} - \mathbf{M}_k = (\mathbf{I} - \mathbf{M}_{k+1}) + (\mathbf{M}_{k+1} - \mathbf{M}_k)$ with orthogonal ranges, so $\|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = \|(\mathbf{I} - \mathbf{M}_{k+1})\boldsymbol{\mu}\|^2 + \sigma^2 \gamma_k$. The difference of the two risks is $\sigma^2(\gamma_k - (p_{k+1} - p_k))$.

19.1.5.3 C. Going deeper

Exercise C1.

Assume normal errors. Compare s_1^2 (short) and s^2 (long) as estimators of σ^2 by mean squared error, when $\boldsymbol{\beta}_2 \neq \mathbf{0}$. Show that s_1^2 is better iff $2(n - p_1) + 4\gamma + \gamma^2 < 2(n - p_1)^2/(n - p)$, and interpret the condition when n is large.